

III. Differences between Flagella & Cilia

Flagella	Cilia
1) Long whip-like	1) Short hair-like
2) One, two, four or many	2) Many
3) Flagella normally originate at the anterior end.	3) Cilia are found all over the body surface or confined to peristomial region.
4) Flagella do not unite to form cirri.	4) In some ciliates, the cilia fuse to produce cirri, or undulating membrane and membranelles,
5) Flagella perform “undular” movement	5) Cilia perform “pendular” movement

5.1.4 Myonemes

They are the contractile fibrils present below the pellicle in the ectoplasm. They occur in flagellates, apicomplexans and ciliates.

5.2 FLAGELLAR AND CILIARY MOVEMENT

5.2.1 Swimming locomotion

This type of locomotion in protozoans is brought about by **flagella** and **cilia**. Flagella and cilia are also called ‘**undulipodia**’ by Hyman. They are useful in bringing about locomotion in water and the body fluid of hosts in the case of a few parasites. Bending movement of a flagellum is brought about by the sliding of microtubules past each other due to the functioning of ‘**dynein arms**’ utilising ATP. A flagellum pushes the

fluid medium at right angles to the point of its attachment, by its bending movement.

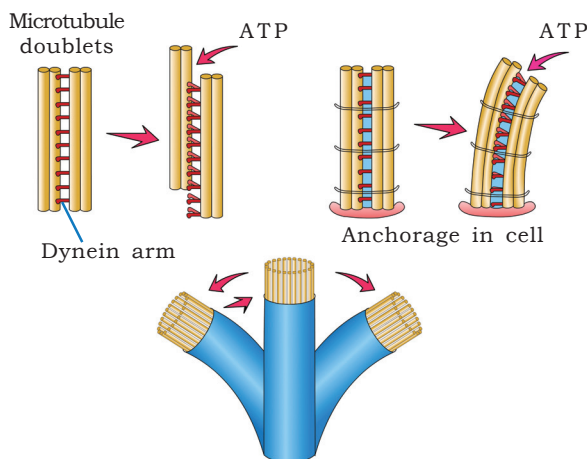


Fig 5.6 Bending of microtubules

Bending movement of flagella and cilia

Dynein arms show a complex cycle of movements using energy provided by ATP (dynein arms are the sites of ATPase activity in the cilia and flagella). The dynein arms of each doublet attach to an adjacent doublet *and pull* the neighbouring doublet. So the doublets slide past each other in opposite directions. The arms release and reattach a little farther on the adjacent doublet and again ‘pull’. As the doublets of a

flagellum or cilium are physically held in place by the radial spokes, the doublets cannot slide past much. Instead, they curve and cause bending of flagellum or cilium. Such bending movements of flagella and cilia play an important role in the flagellar and ciliary locomotion.

If, nexins and radial spokes of an axoneme in a flagellum/cilium are subjected to enzymatic digestion and provided with ATP, what happens to the doublets?

5.2.2 Flagellar locomotion

Flagellum shows undulation movements, sidewise lash movements and spiral gyration movements.

- a) **Undulation movements:** Undulatory movements from the base to the tip causes a **pushing force** (like the propeller of a boat) and the organism is *pushed* backwards (undular movement). Undulation from the tip to the base causes a **pulling force** (like the propeller of an aeroplane) and the organism is *pulled* forwards. If the flagellum bends to one side and shows a wave like movement from the base to the tip, the organism moves laterally in the opposite direction.
- b) **Sidewise-lash movement:** The flagellar movement of many organisms is a sidewise-lash which consists of two 'strokes' namely the **effective** or **propulsive stroke** and the **recovery stroke**.
 - (i) **Effective stroke:** Flagellum becomes rigid and starts bending to one side beating against the water. This beating against water is at right angles to the body axis and the organism moves forwards (the direction of movement is parallel to the surface of the body).
 - (ii) **Recovery stroke:** Flagellum becomes comparatively soft so as to offer least resistance to water and moves backwards to its original position. It is called 'recovery stroke'.
- c) **Simple conical gyration movements:** In simple conical gyration movement, a flagellum turns like a screw. This exerts the propelling action that pulls the organism forwards through water with a spiral **rotation** around the axis of movement and **gyration** on its own axis.

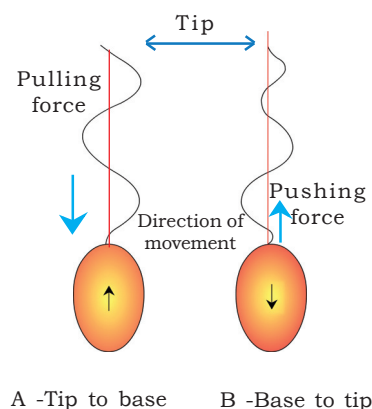


Fig. 5.7 Undular movement of a flagellum

5.2.3 Ciliary Locomotion

Ciliary locomotion is observed in ciliates. It is performed by bending movements of the cilia. Cilium also shows swift back and forth movements during locomotion as in the case of a flagellum. These are called *effective* and *recovery* strokes respectively. Cilium moves the water parallel to the surface of its attachment like that of a *paddle* or *pendular movement*. The movement of water may also be described as perpendicular to the axis of cilium.

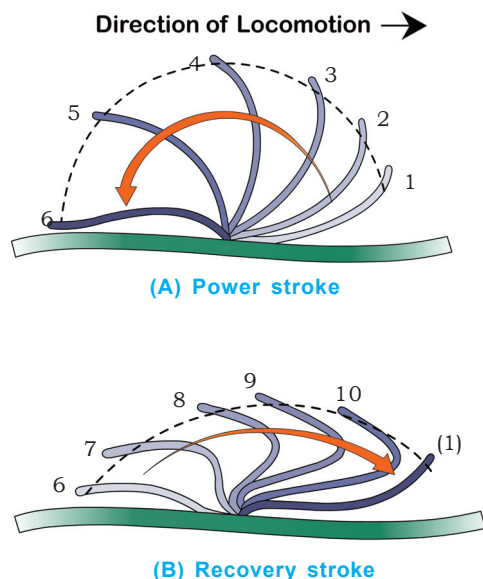


Fig. 5.8 (A) Effective stroke
(B) Recovery stroke

- a) **Effective stroke:** Cilium bends backwards and beats against the water. It is called the 'effective stroke'. Body moves forwards in this stroke, as cilium pushes the water backwards. Cilium moves like *pendulum* or a *paddle*. (*pendular movement*)
- b) **Recovery stroke:** The cilium after the effective stroke, comes to its original position by its backward movement without any resistance. It is called 'recovery stroke'.

In **Paramecium** cilia show two types of coordinated movements.

- a. **Synchronous movement:** Cilia in a transverse row beat simultaneously in one direction. It is called **synchronous movement**.
- b. **Metachronous movement:** The sequential movement of cilia, in a longitudinal row, one after the other in one direction is called **metachronous**

movement. This movement passes like a wave. It resembles the movement of plants in a paddy field due to the blow of wind from one direction to the other. The ciliary locomotion is coordinated by the *infraciliary system* through a neuromotor centre called **motorium**, present near the cytopharynx. Ciliary locomotion is faster than flagellar locomotion. Hence, ciliates are the fastest protozoans.

Flagella are responsible for the movement of the organism, whereas cilia bring out movement of either the organism or the surrounding fluid as seen in ciliary mode of feeding in certain organisms.

5.2.4 Gliding Locomotion

The small zig-zag movement in some protozoans caused by the contraction and relaxation of myonemes present below the pellicle in the ectoplasm is called **gliding locomotion**. **Myonemes** are contractile fibrils which are similar to the myofibrils of higher protozoans. e.g. ciliates, flagellates etc.

Is it correct to say 'all undulipodia' do not show undular movements?